

Data Science in Medical Imaging

Coursework Minor A2

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Abstract

This report accompanies the submitted code and experiments for the three coursework modules: tomographic reconstruction, MRI denoising, and a written critical comparison of segmentation methods. Quantitative metrics are reported alongside representative visual outputs, and parameter settings are stated so results can be reproduced from a clean checkout.

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1 Reproducibility notes

- Code lives in `med_im/`. Experiments are run from `notebooks/` (and any scripts referenced there).
- Figures and tables used in this report should be generated by tracked code; each figure caption states which notebook/script produced it.
- Random seeds and key parameters (noise level, number of projections, angular range, filter settings) are stated next to results.

2 Module 1: Tomographic reconstruction

2.1 Exercise 1.1: Reducing dose

Setup

The CT slice provided in the coursework was loaded from `data/CT_exercise_1.png` and converted to grayscale. The working image size was 512×512 pixels with intensity range approximately $[0, 10^{-3}]$.

Noise simulation

Noisy sinograms were simulated by combining Poisson noise on the transmitted intensity with additive Gaussian noise on top. The Gaussian component used $\mu = 0$ and $\sigma = 0.05$ while the Poisson component used three incident fluence settings $I_0 \in \{10^5, 10^3, 10^2\}$. Each condition was simulated at three projection counts $N_\theta \in \{360, 90, 20\}$.

Reconstructions: FBP vs gradient descent (SIRT)

Each noisy sinogram was reconstructed using filtered back-projection (FBP, ramp/Ram-Lak filter) and a simple gradient descent / SIRT-style update using repeated backprojection of the residual.

Clinical interpretation

Reducing I_0 (dose) and reducing the number of angles both degraded reconstruction quality; in the lowest dose / sparse-angle settings the reconstructions show more visible artefacts and loss of fine structure, which would plausibly make small lesions or subtle edges harder to detect. In practice there is a trade-off between radiation exposure, acquisition time, and diagnostic reliability, and iterative methods may partially compensate for reduced data but cannot fully recover missing information.

How to get better images

Two straightforward ways to improve quality in these settings are to add explicit regularisation (for example, total variation or edge-preserving penalties) and to use more accurate statistical models for the noise (e.g. Poisson likelihood rather than a simple least-squares residual). A third option is to use learned priors or plug-and-play denoisers inside an iterative reconstruction loop, although this requires careful validation.

2.2 Exercise 1.2: Limited angle tomography

Setup and simulations

The same dose and projection-count experiment from Exercise 1.1 was repeated using limited angular coverage. Three angular ranges were tested: 180°, 120°, and 40°.

Observed differences and explanation

As the angular range shrinks, the reconstruction becomes increasingly anisotropic: some edges are recovered reasonably while others smear into streaks. This is consistent with the Fourier slice theorem intuition: limited angle sampling leaves regions of frequency space unmeasured (a “missing wedge”), so features aligned with those missing frequencies are poorly constrained.

2.3 Exercise 1.3: Reconstruction

Alternative FBP filters

Two alternative filters tested here were:

- Shepp–Logan: a ramp filter multiplied by a sinc window, which rolls off very high frequencies and can reduce noise while keeping edges fairly sharp.
- Cosine: a ramp filter multiplied by a cosine window, which suppresses high frequencies more strongly; this tends to look smoother but also more blurred.

OS-SART / ordered subsets vs SIRT

Ordered subsets updates (OS-SART) were implemented by splitting projection angles into subsets and updating the image multiple times per iteration. In this small experiment OS-SART gives a similar qualitative reconstruction to full-batch SIRT but with different convergence behaviour depending on subset size and step size.

PET vs CT iterative methods

3 Module 2: MRI image denoising

3.1 Exercise 2.1: Visualisation and identifying noise

k-space and coil dimension

The k-space data (`data/knee.npy`) was loaded as a 3D array with the coil dimension on axis 0 (six coils). Log-magnitude k-space was plotted for each coil.

Image-space magnitude and phase

Each coil was transformed to image space using an inverse FFT. For one coil, both magnitude and phase images were visualised, then magnitude images were shown across all coils.

Coil combination

Coils were combined using root-sum-of-squares (RSS), $\sqrt{\sum_c |I_c|^2}$, producing a single magnitude image with improved SNR compared to single-coil images.

3.2 Exercise 2.2: Removing noise

Three image-space denoisers

Three denoising methods were applied coil-wise in image space (on magnitude images): Gaussian filtering, bilateral filtering, and wavelet denoising.

Brief comments (matching the development notebook):

- Gaussian: smooths uniformly; reduces noise but blurs edges.
- Bilateral: more edge-preserving; smoother flat regions while keeping sharper boundaries.
- Wavelet: removes noise-like coefficients across scales; can preserve edges but depends on thresholding.

k-space Butterworth low-pass

A Butterworth low-pass filter was applied directly in k-space for the first coil before inverse FFT. This reduces high-frequency noise, but if the cutoff is too aggressive it can blur detail and may introduce ringing.

Denoising on combined image

Finally, a mild Gaussian filter was applied to the RSS combined image as a simple post-processing step. Stronger filtering would reduce noise further but blur edges.

4 Module 3: Image segmentation methods (written component)

Note: The substantive written content for this module must be authored without generative AI assistance. The structure below is provided as a template only.

4.1 Paper A (processing-based / non-ML)

4.2 Paper B (ML/DL)

4.3 Critical comparison (max one page)

4.4 Proposed improvement (max half page)

References

References

A Optional: Included supporting PDFs

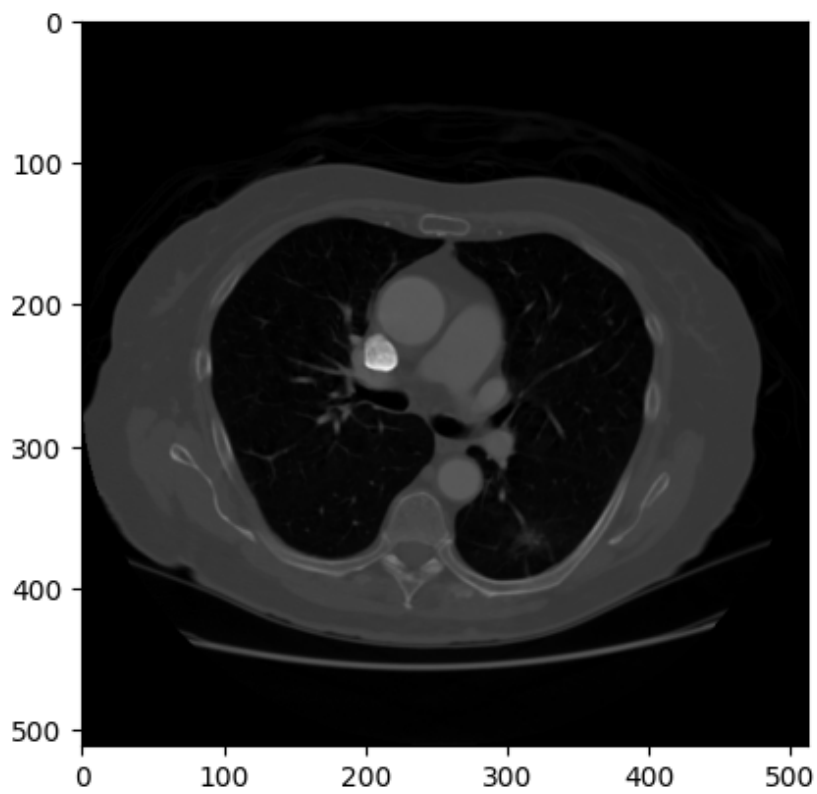


Figure 1: Input CT image used for Module 1 (generated in `notebooks/development.ipynb`).

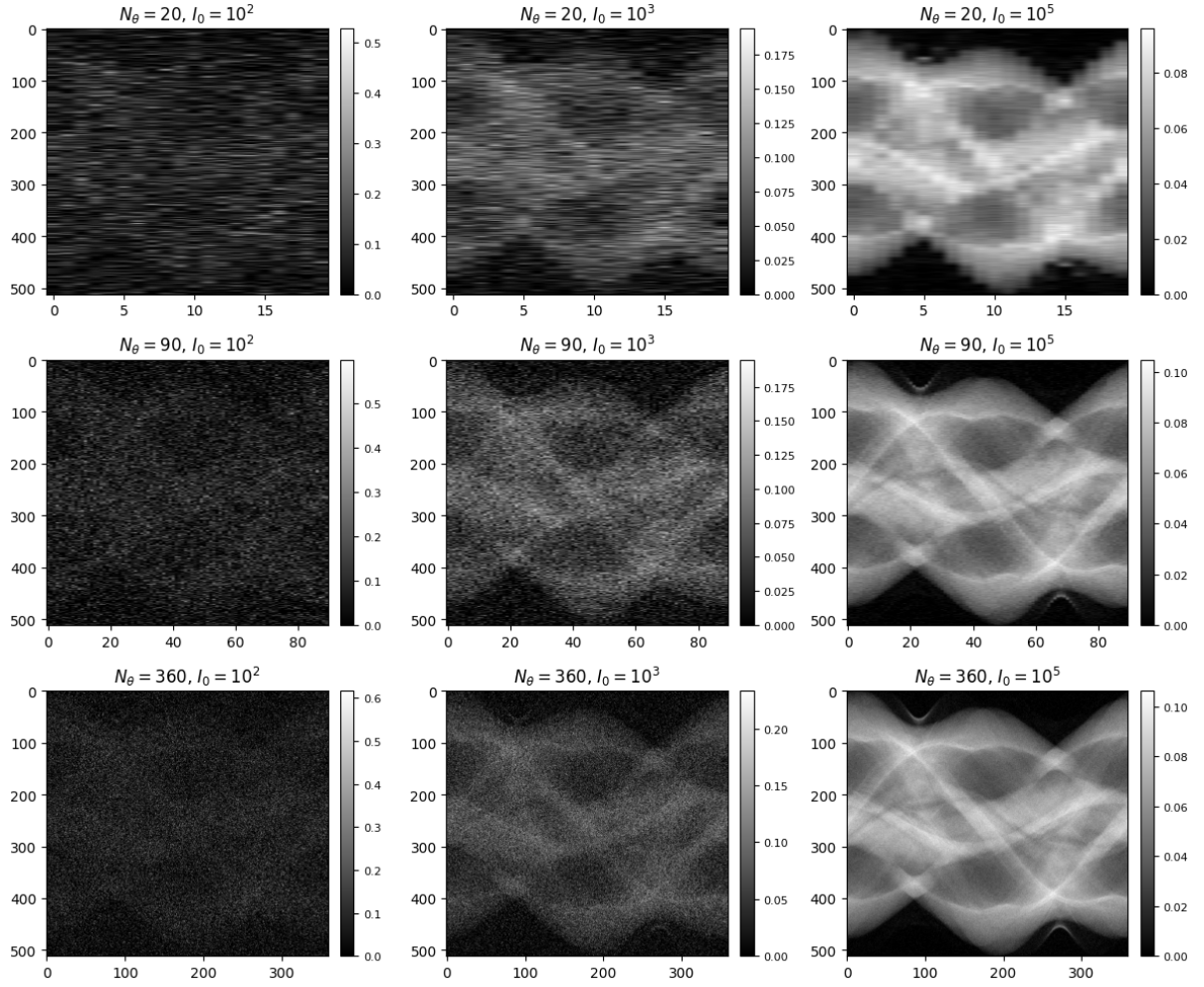


Figure 2: Noisy sinograms across projection count (N_θ) and dose setting (I_0) (generated in `notebooks/development.ipynb`).

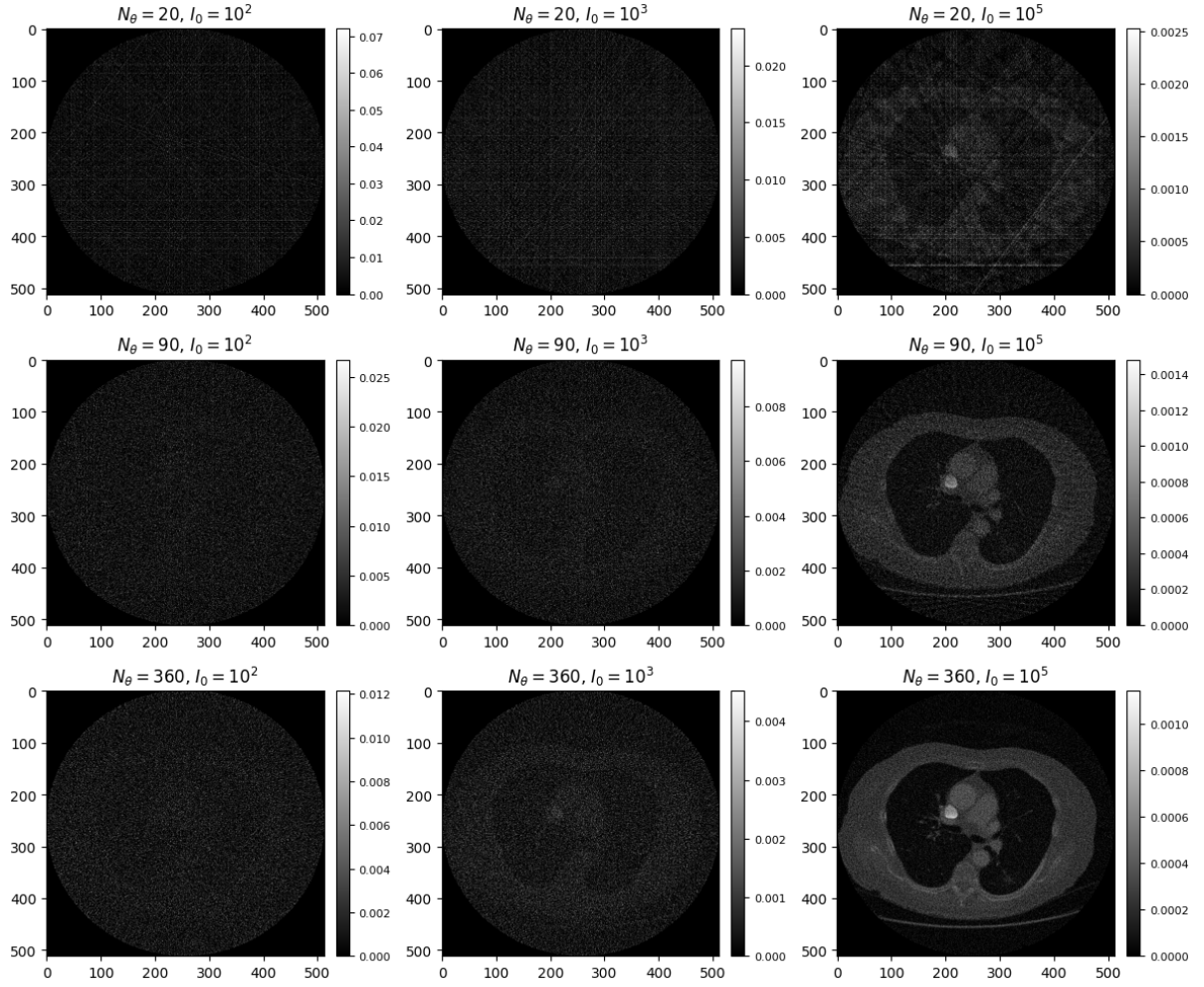


Figure 3: FBP reconstructions across N_θ and I_0 (generated in `notebooks/development.ipynb`).

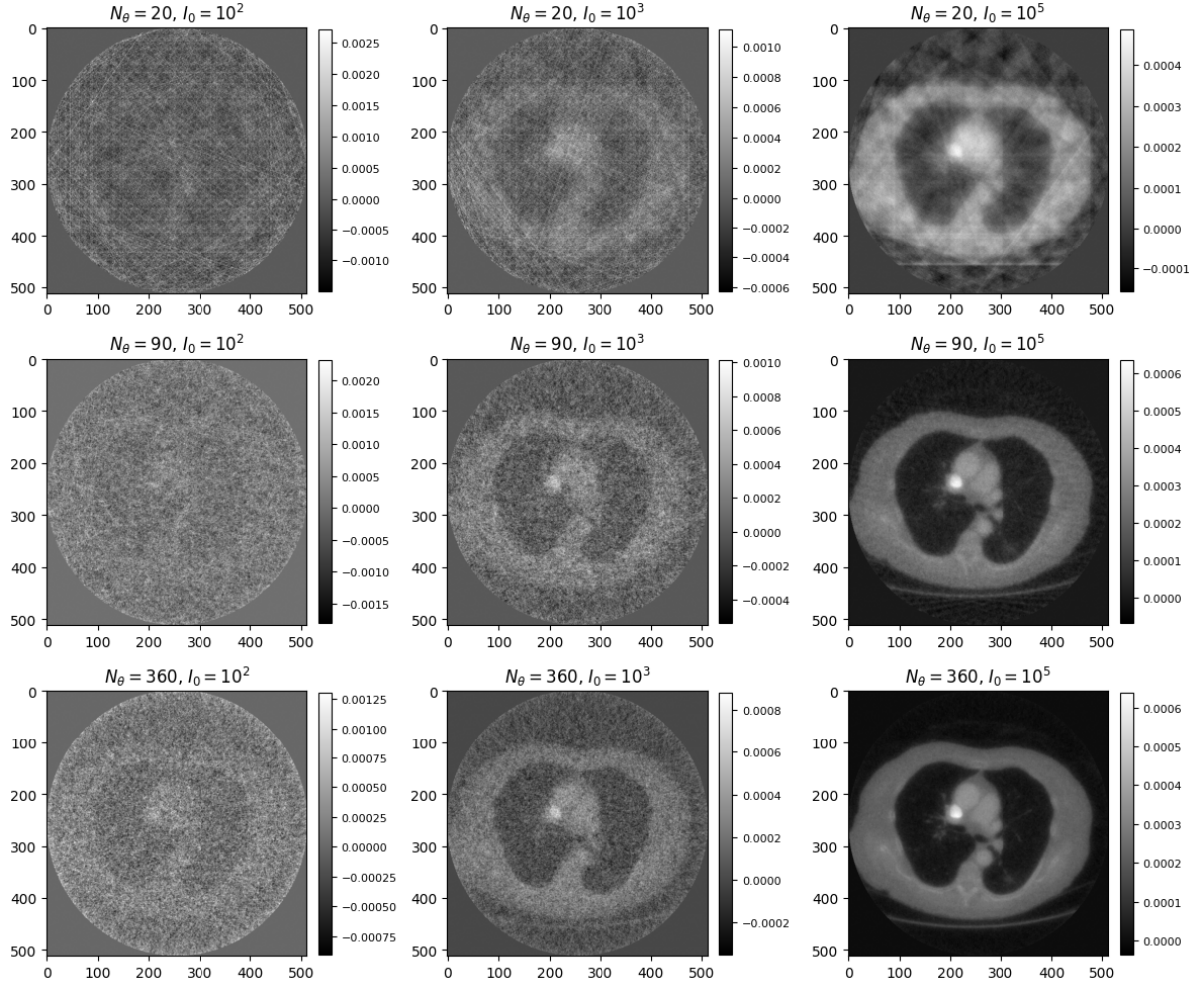


Figure 4: Gradient descent / SIRT reconstructions across N_θ and I_0 (generated in `notebooks/development.ipynb`).

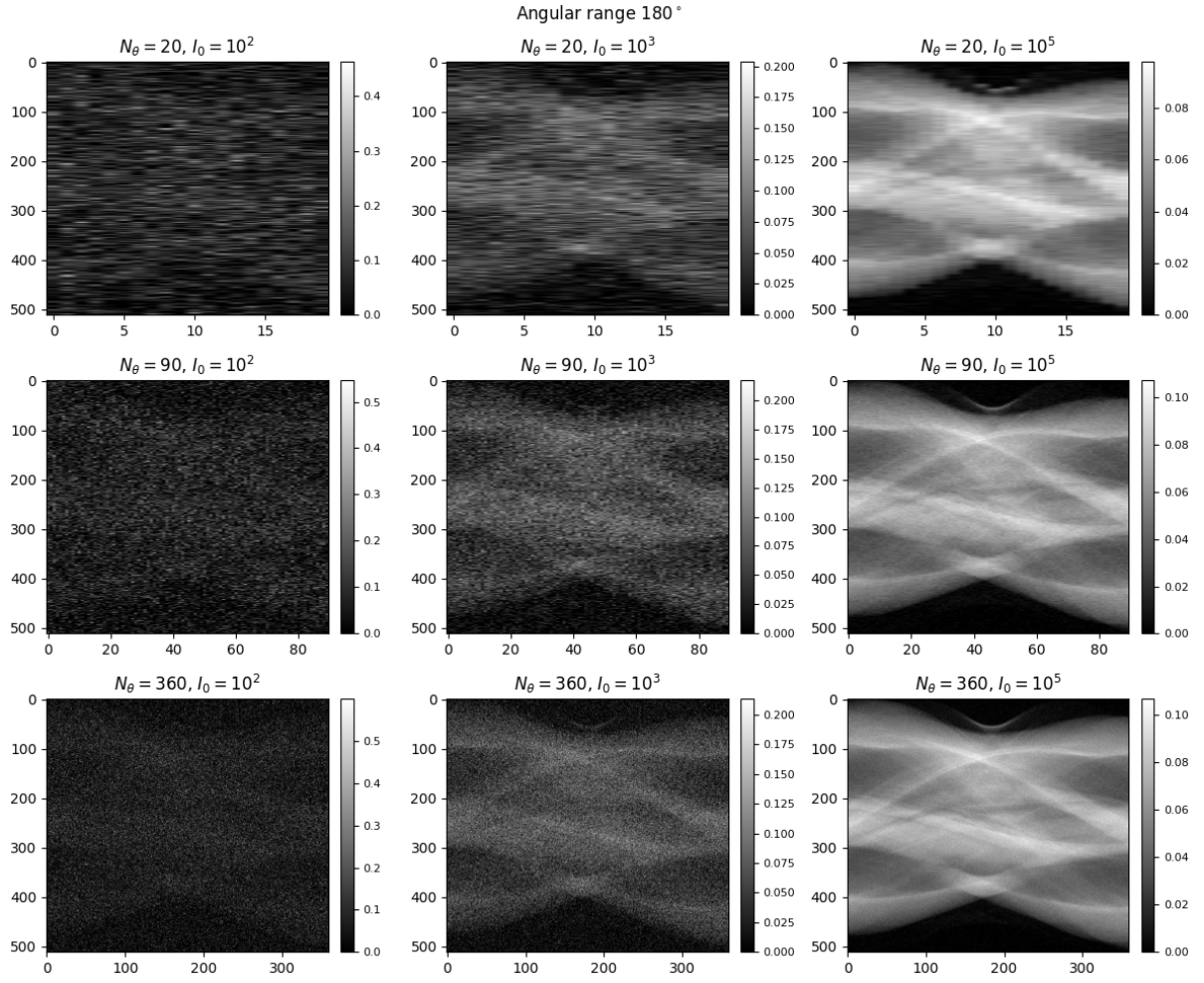


Figure 5: Noisy sinograms for angular range 180° (generated in notebooks/development.ipynb).

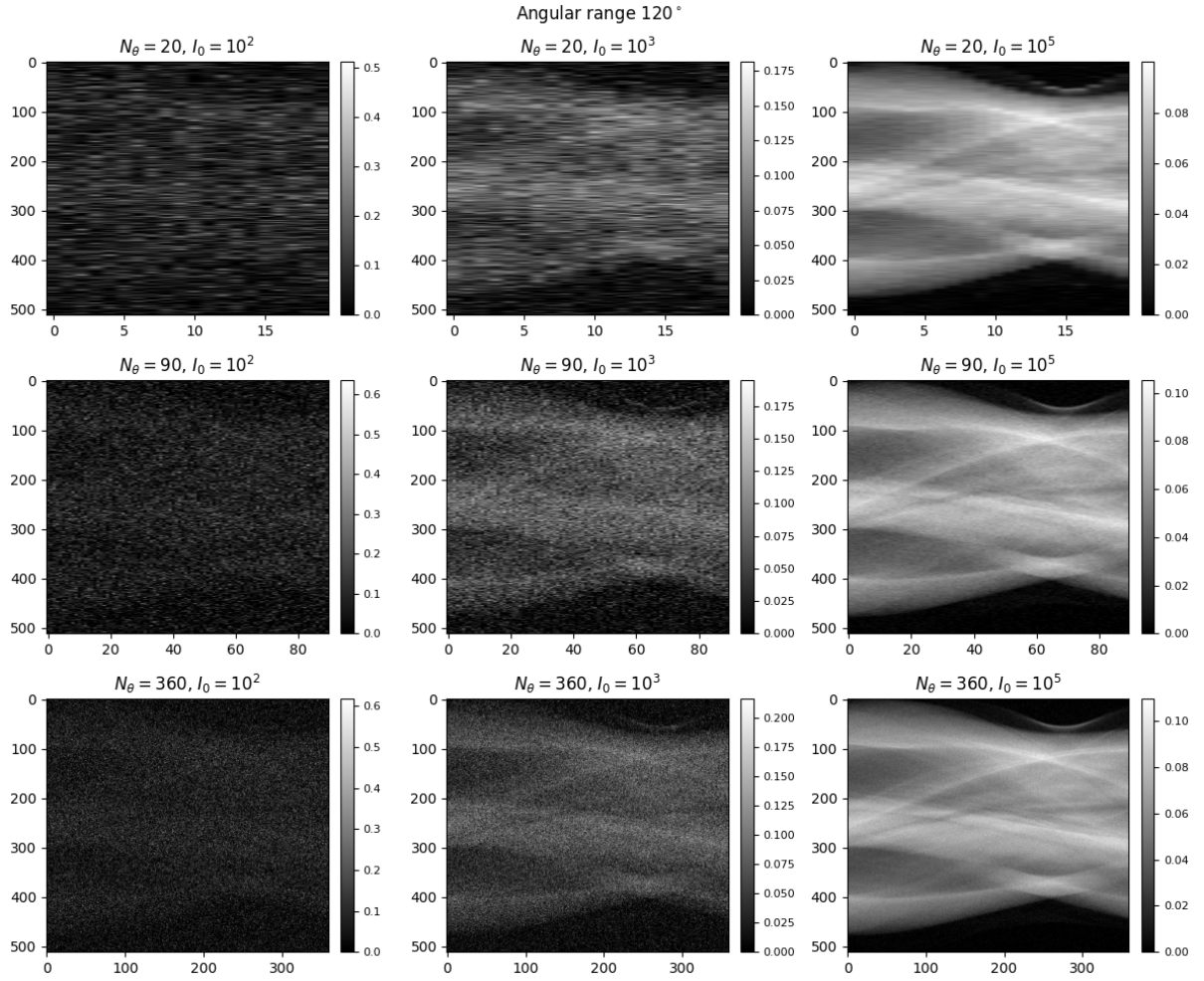


Figure 6: Noisy sinograms for angular range 120° (generated in `notebooks/development.ipynb`).

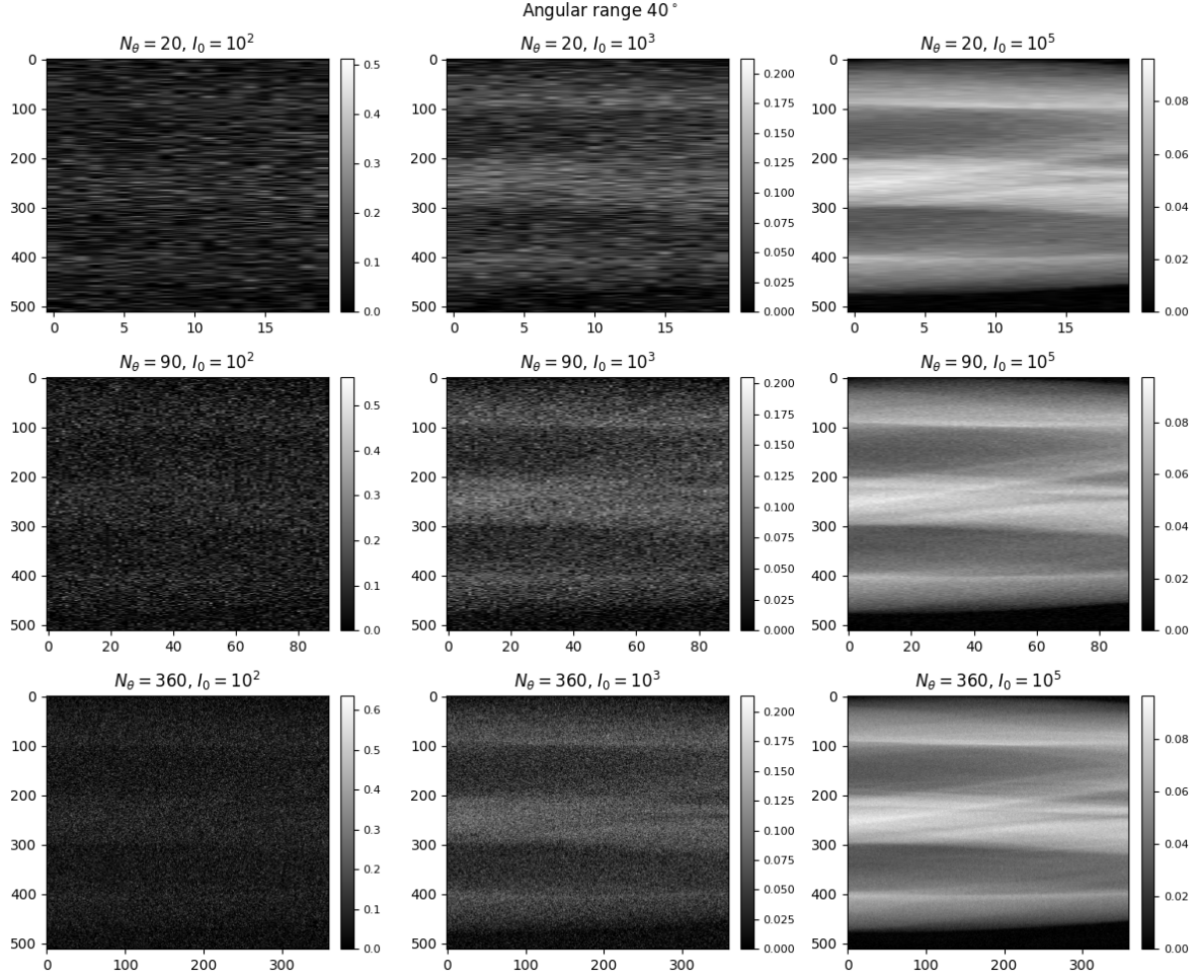


Figure 7: Noisy sinograms for angular range 40° (generated in `notebooks/development.ipynb`).

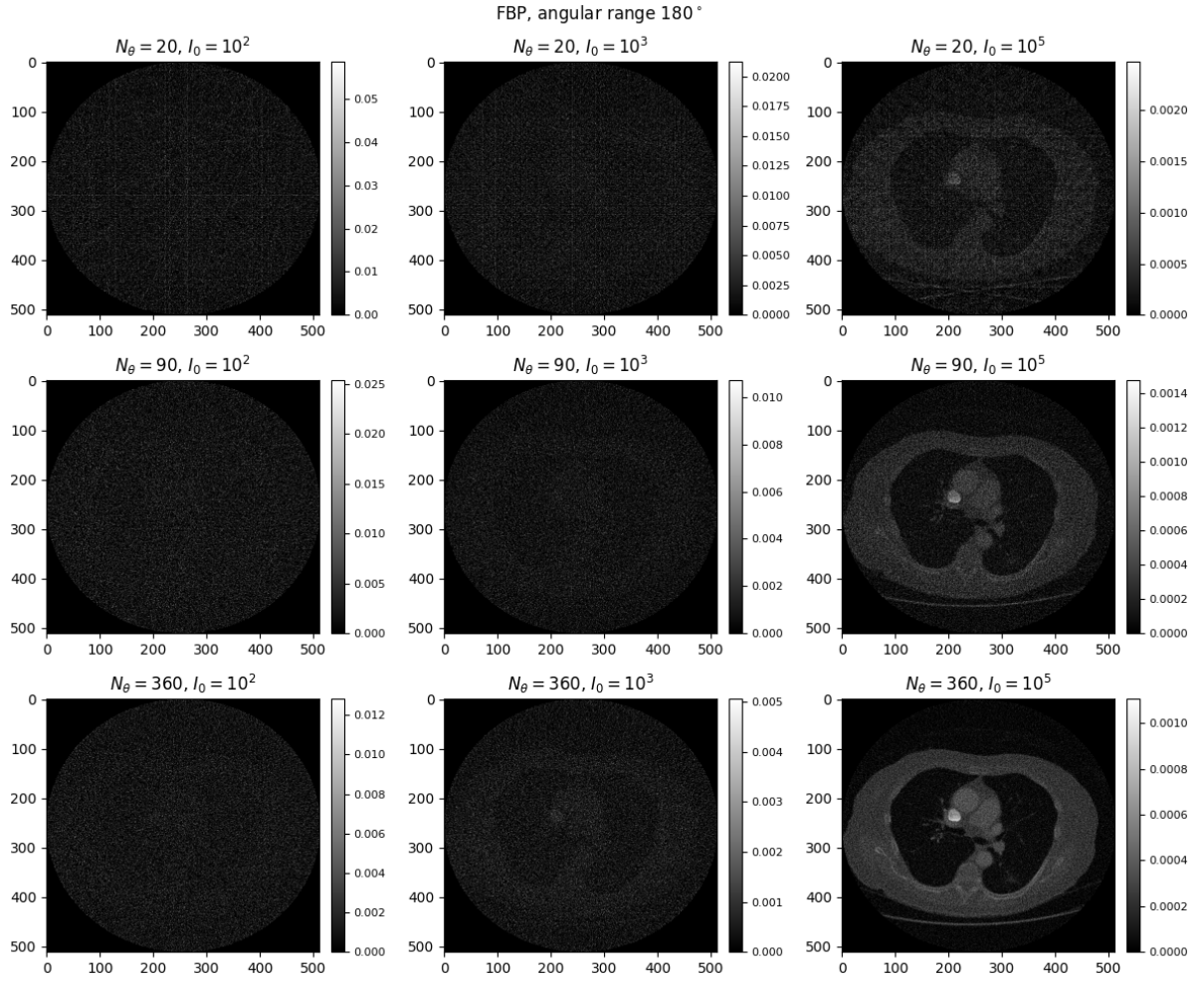


Figure 8: FBP reconstructions at 180° angular range (generated in notebooks/development.ipynb).

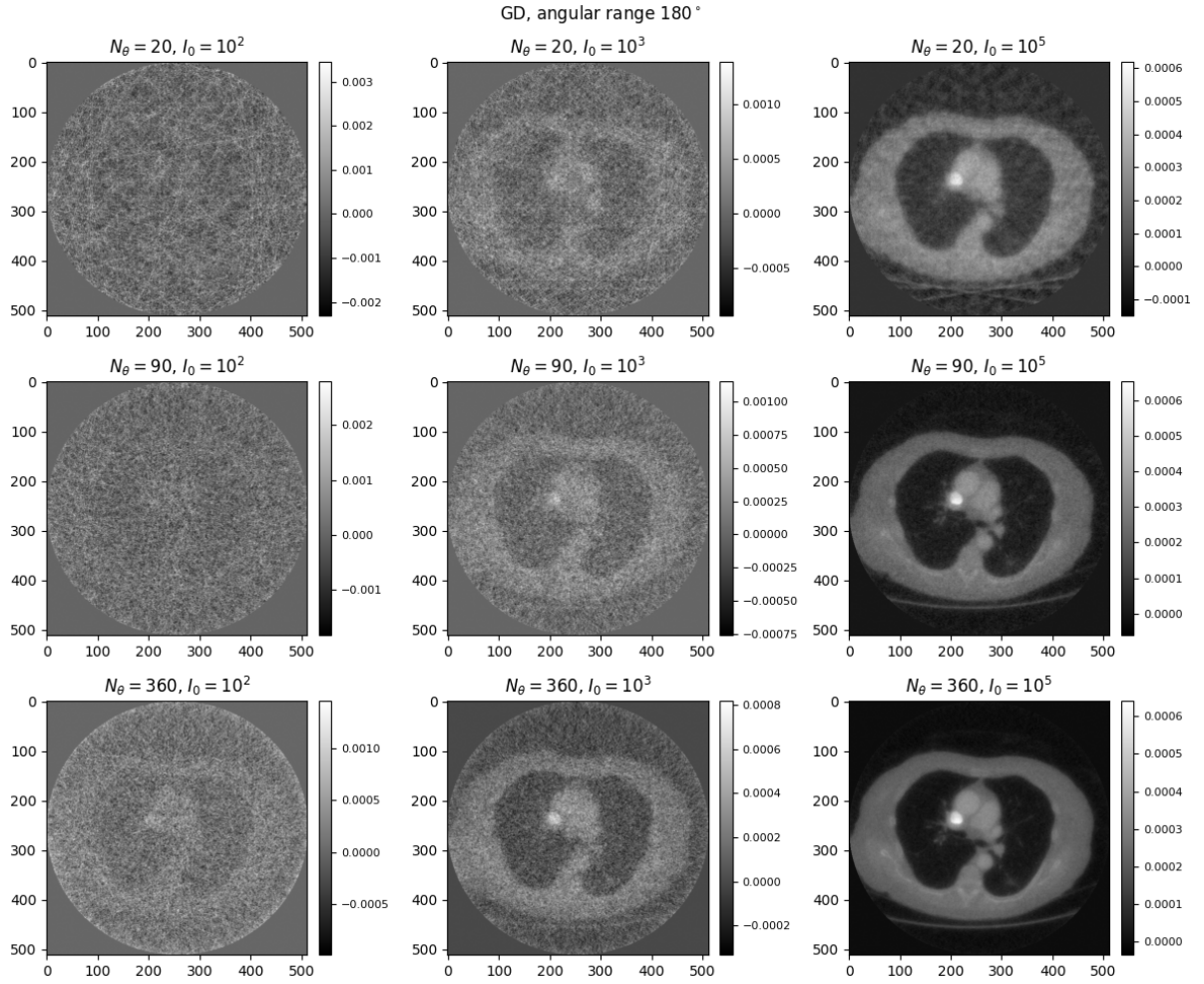


Figure 9: GD/SIRT reconstructions at 180° angular range (generated in notebooks/development.ipynb).

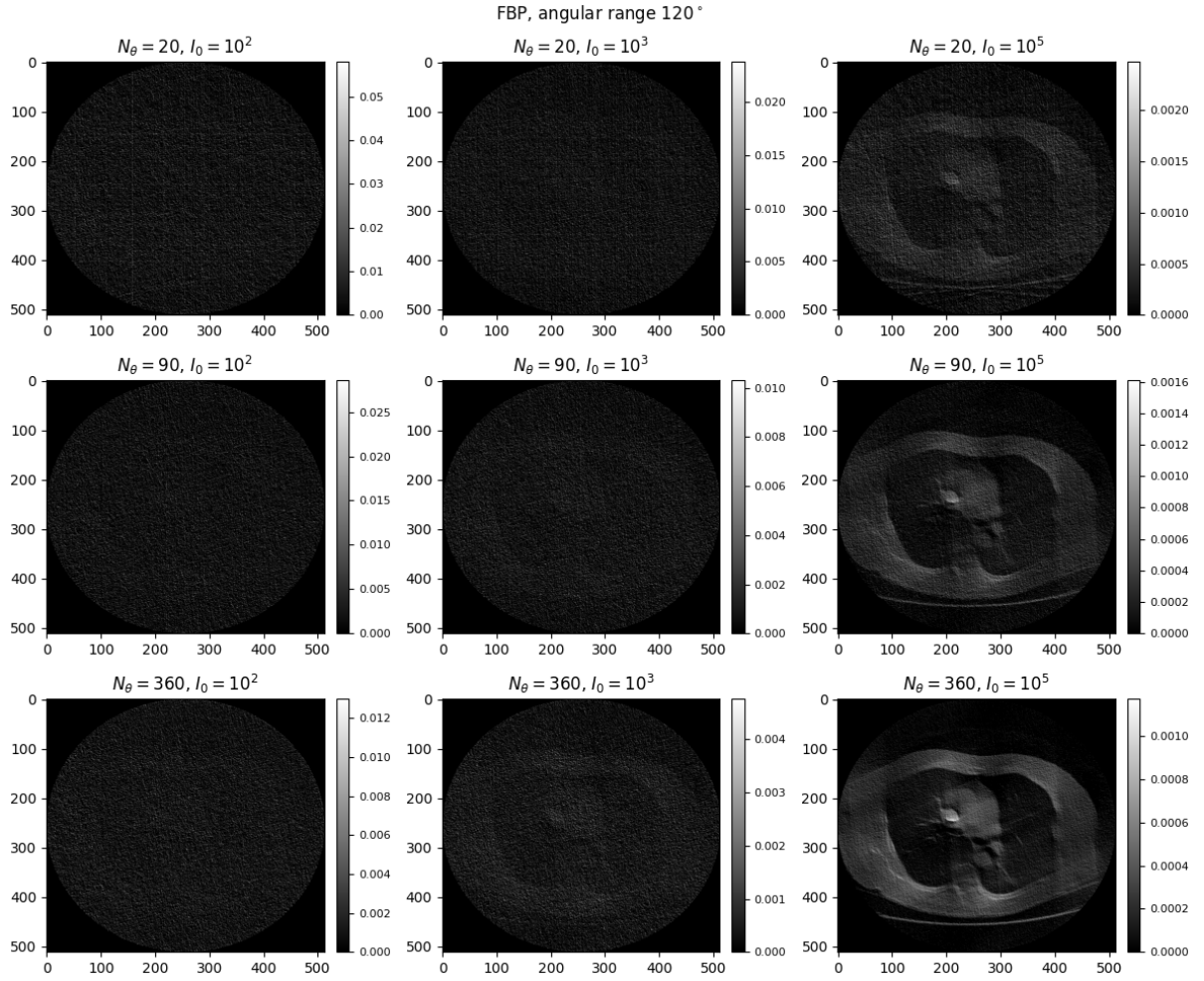


Figure 10: FBP reconstructions at 120° angular range (generated in `notebooks/development.ipynb`).

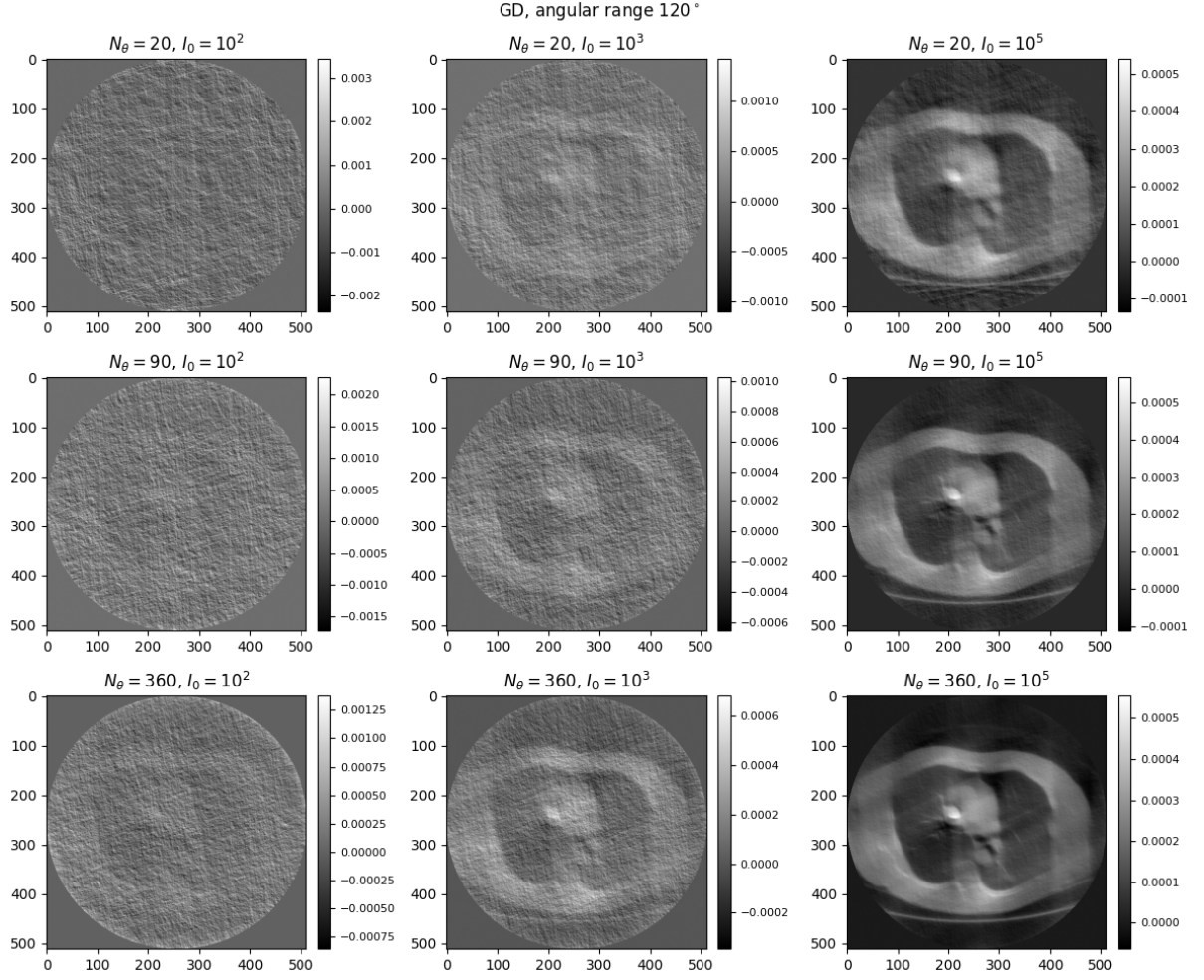


Figure 11: GD/SIRT reconstructions at 120° angular range (generated in notebooks/development.ipynb).

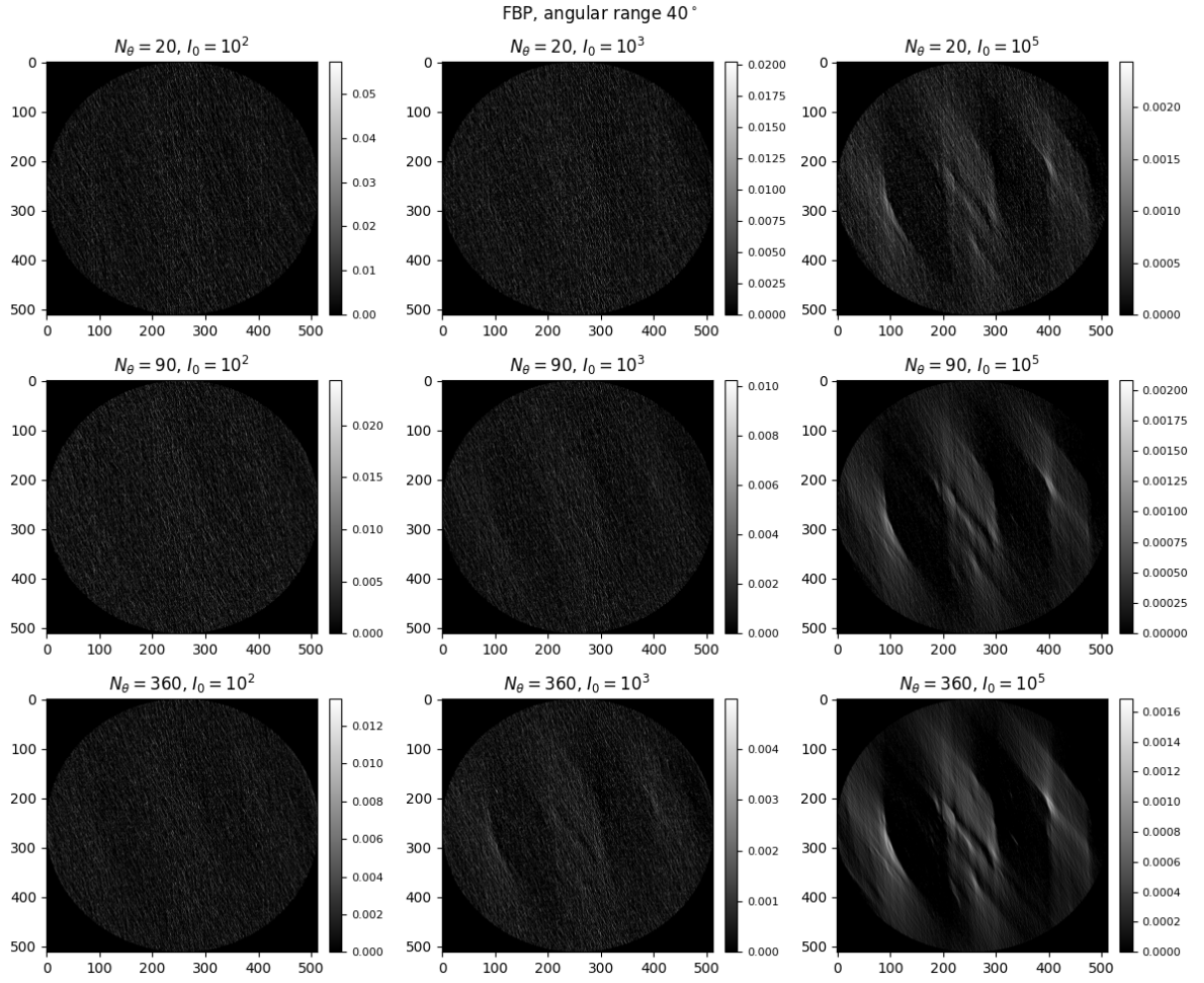


Figure 12: FBP reconstructions at 40° angular range (generated in notebooks/development.ipynb).

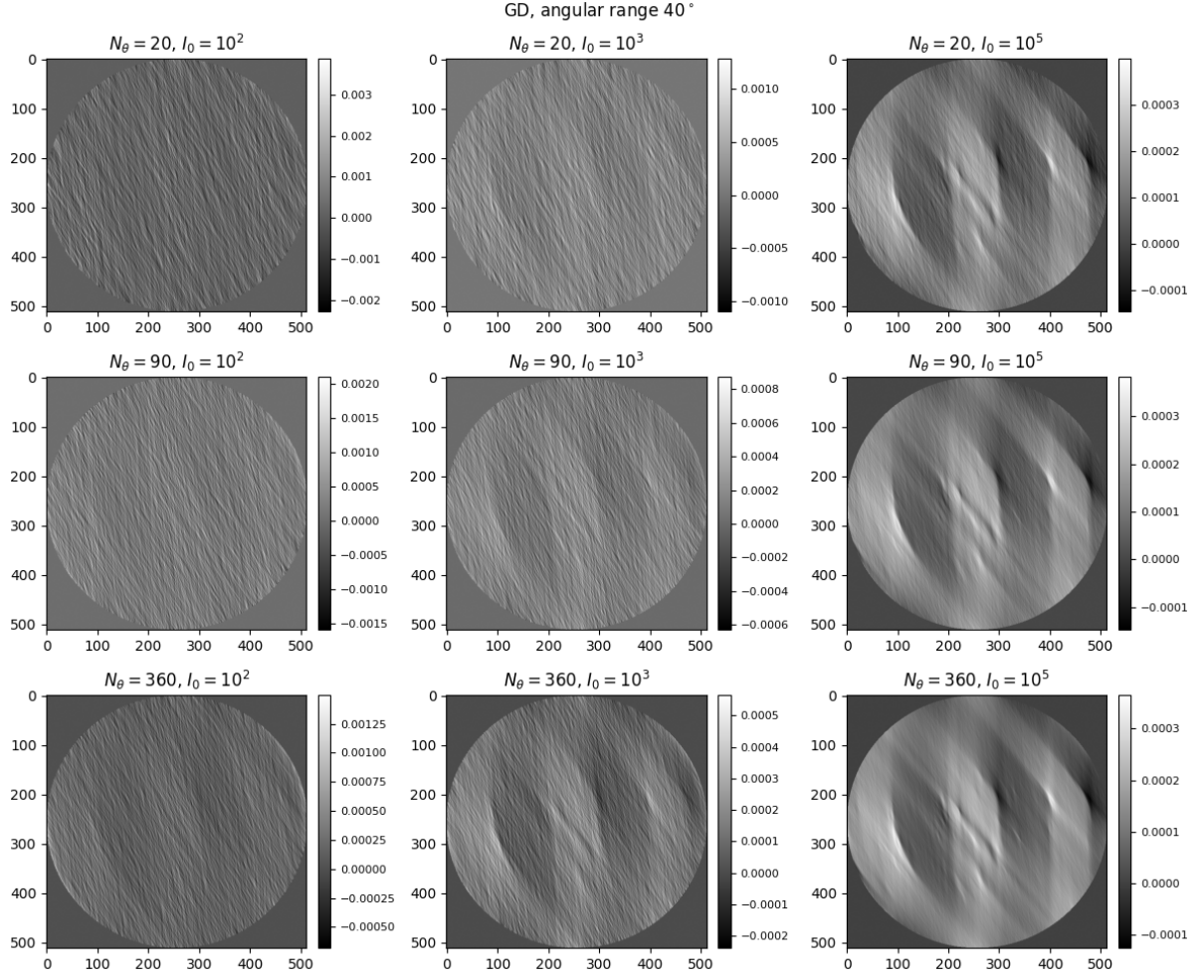


Figure 13: GD/SIRT reconstructions at 40° angular range (generated in `notebooks/development.ipynb`).

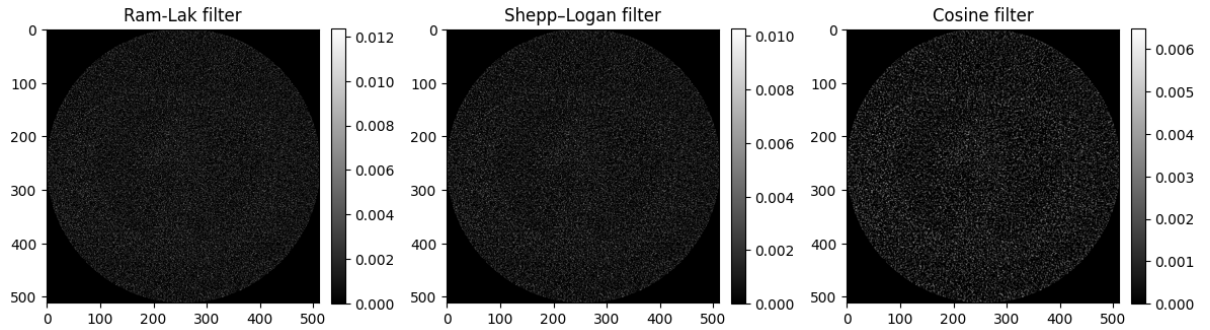


Figure 14: Filter comparison (Ram-Lak / Shepp-Logan / Cosine) on a low-dose example (generated in `notebooks/development.ipynb`).

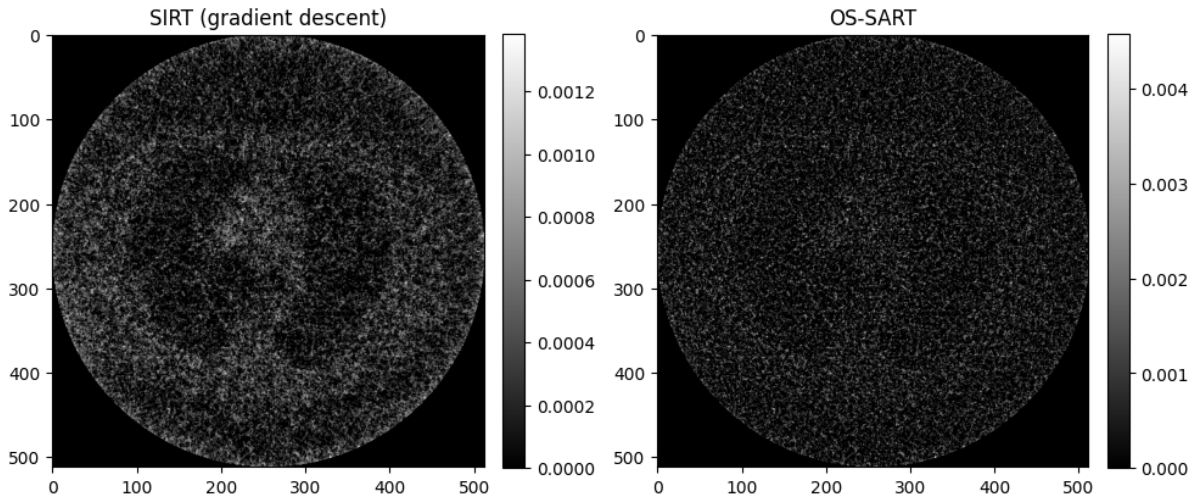


Figure 15: Comparison of SIRT (gradient descent) vs OS-SART for the same acquisition setting (generated in `notebooks/development.ipynb`).

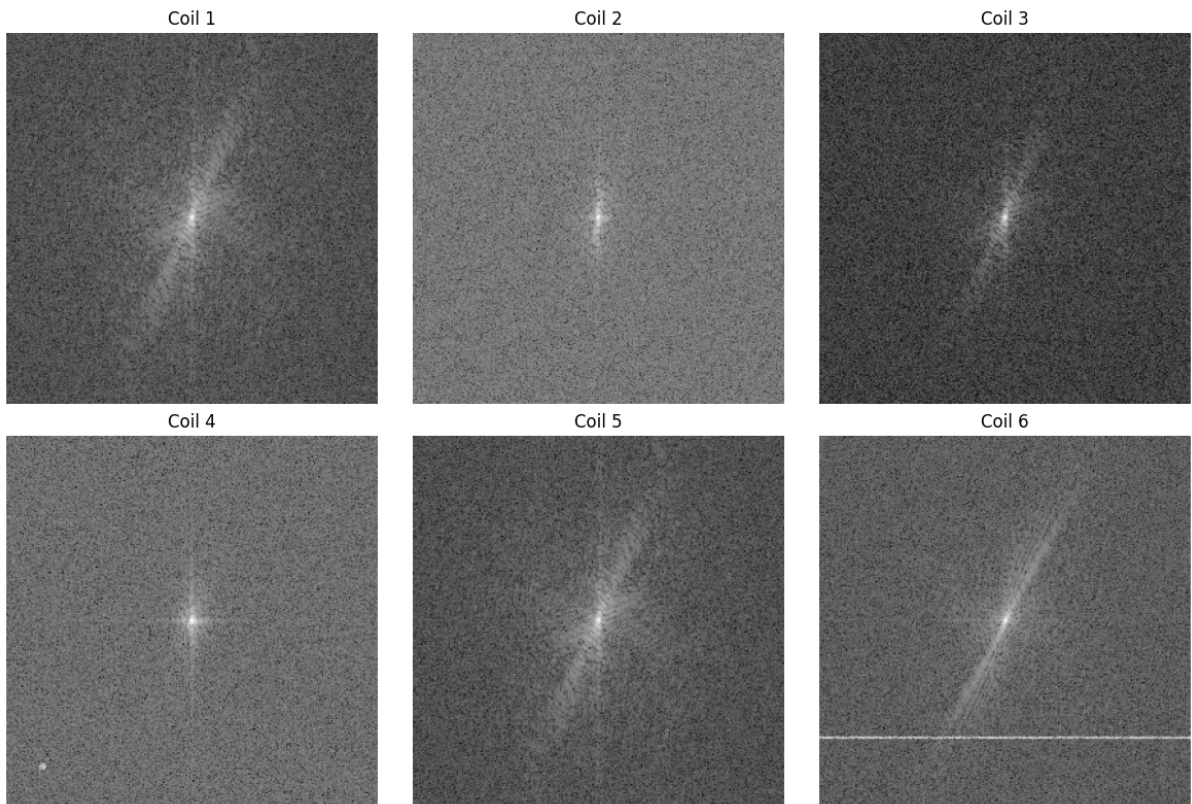


Figure 16: Log-magnitude k-space for each coil (generated in `notebooks/development.ipynb`).

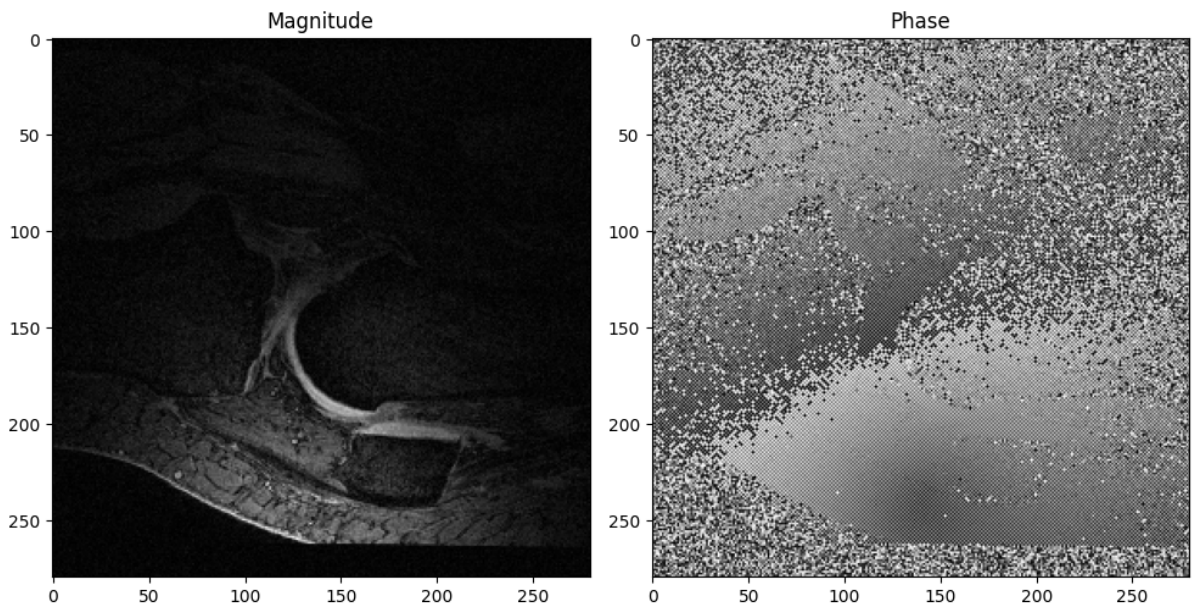


Figure 17: Magnitude and phase image from one coil after inverse FFT (generated in `notebooks/development.ipynb`).

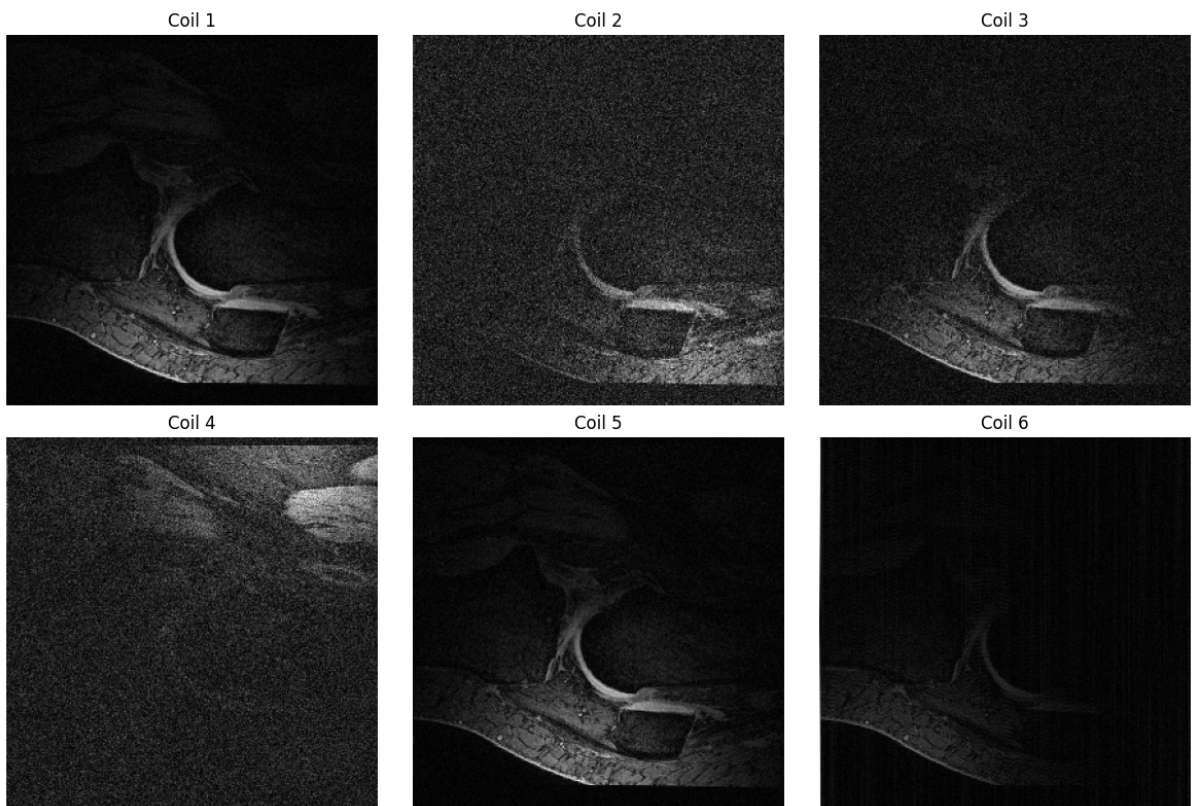


Figure 18: Magnitude images in image space across all coils (generated in `notebooks/development.ipynb`).

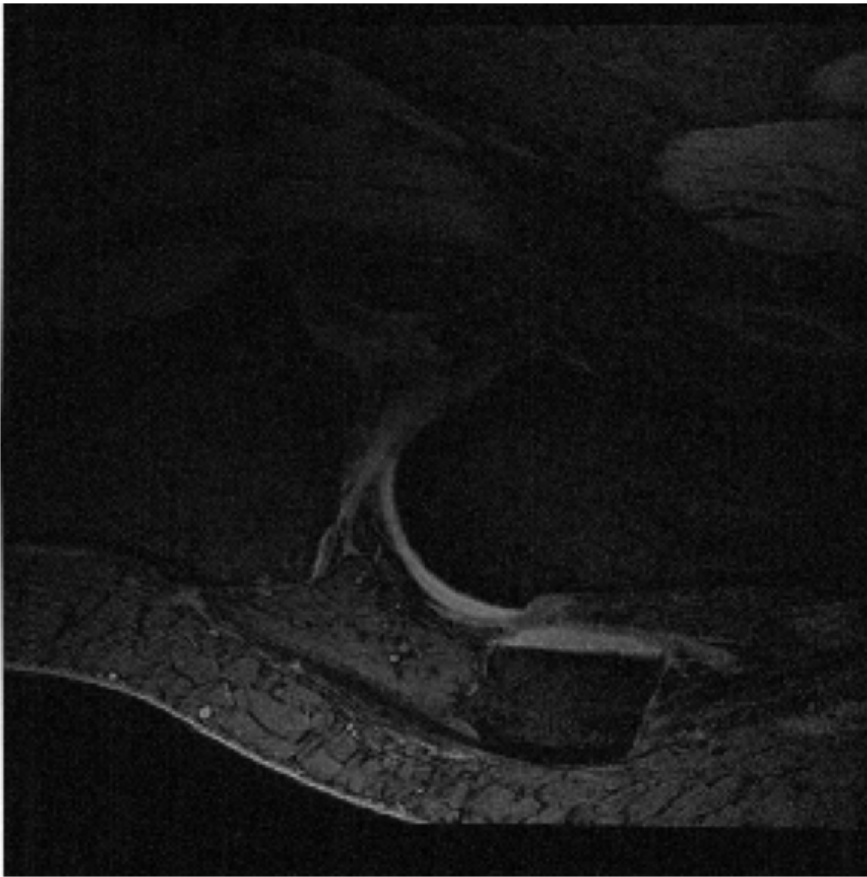


Figure 19: RSS combined magnitude image (generated in `notebooks/development.ipynb`).

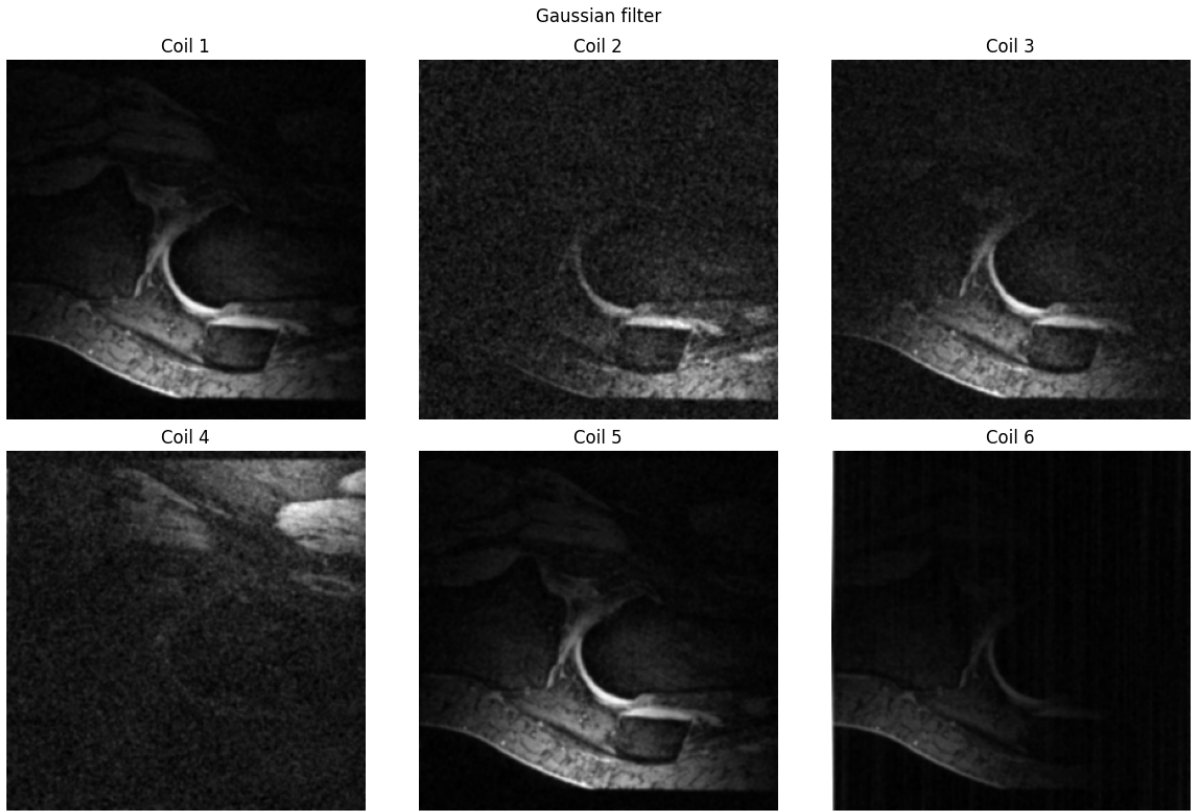


Figure 20: Gaussian denoising applied per coil (generated in `notebooks/development.ipynb`).

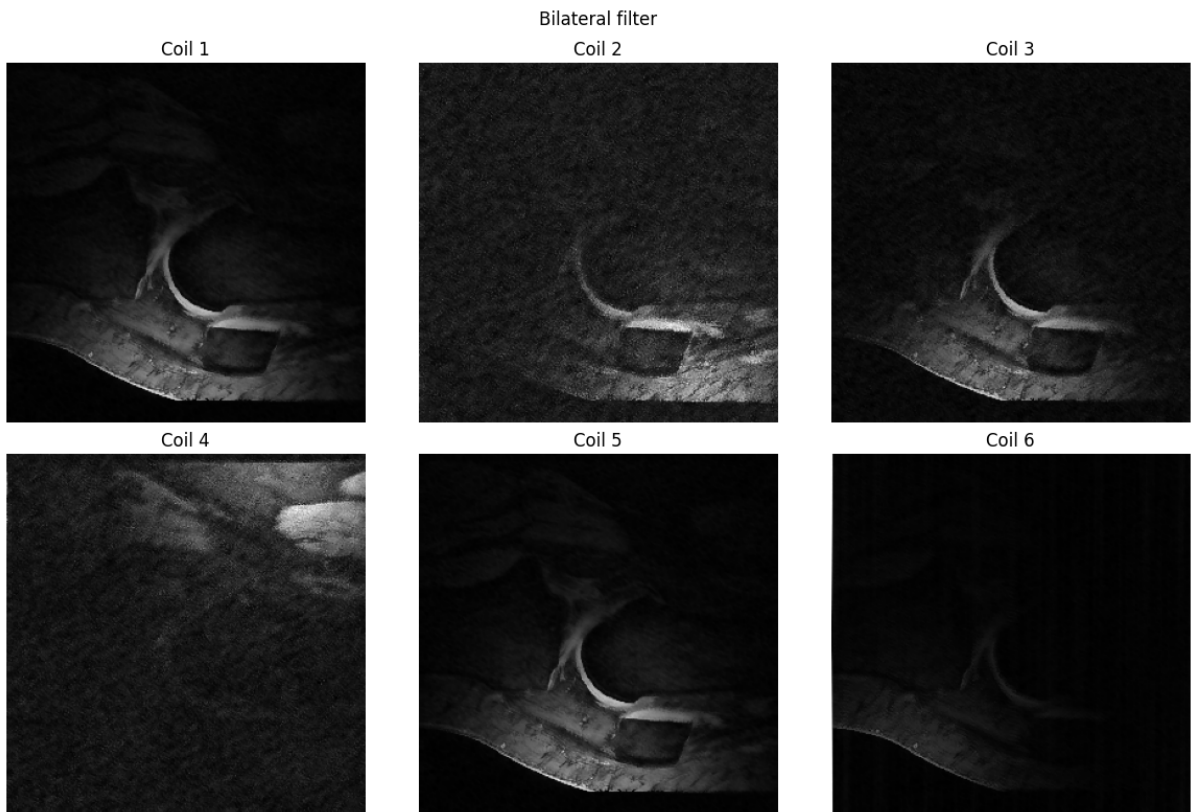


Figure 21: Bilateral denoising applied per coil (generated in `notebooks/development.ipynb`).

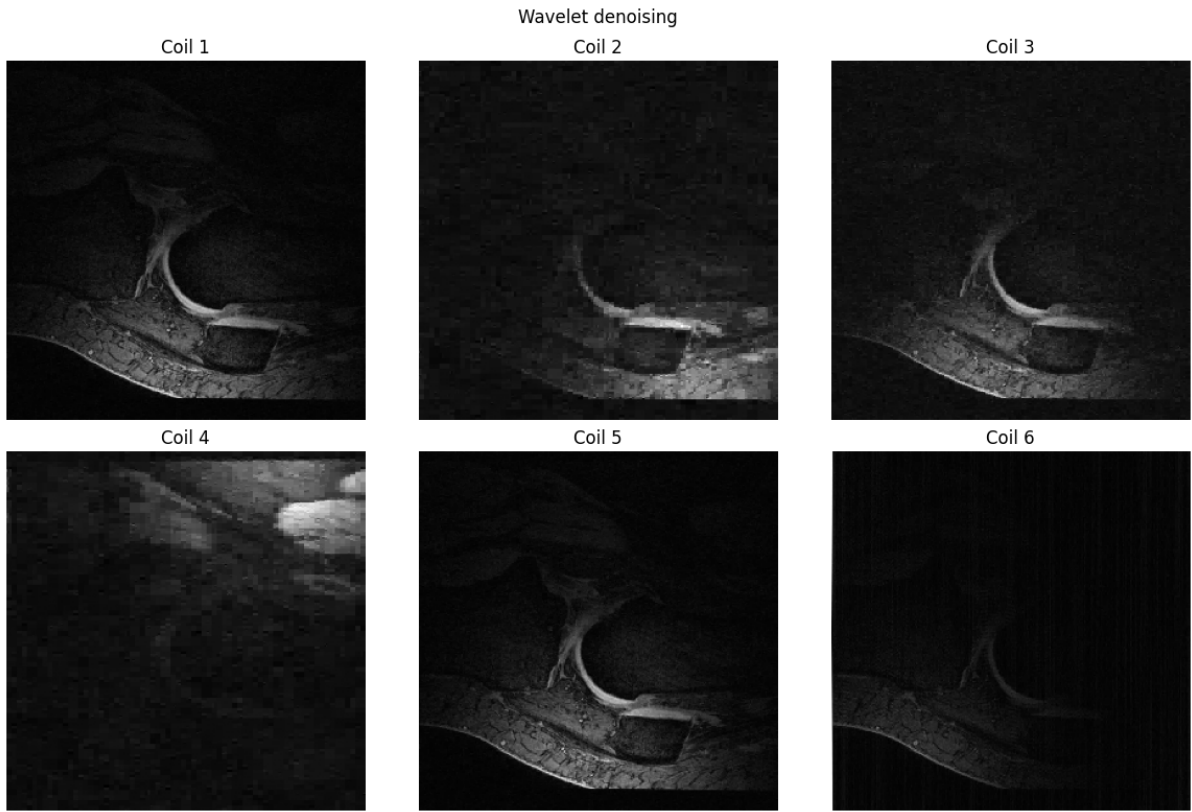


Figure 22: Wavelet denoising applied per coil (generated in `notebooks/development.ipynb`).

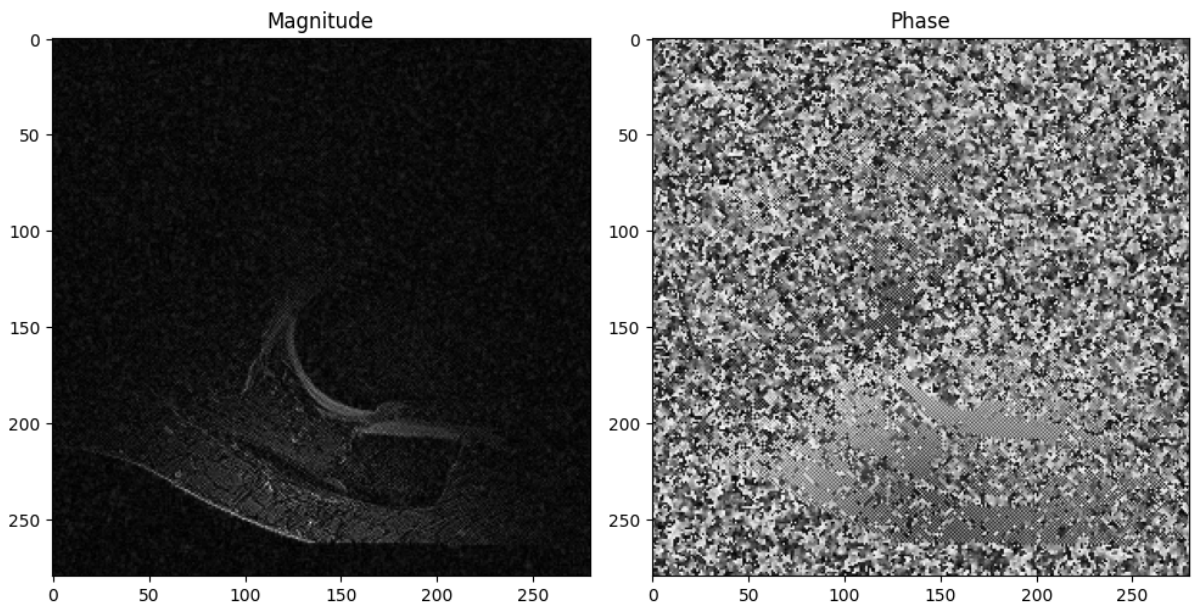


Figure 23: First-coil magnitude and phase after Butterworth low-pass in k-space (generated in `notebooks/development.ipynb`).

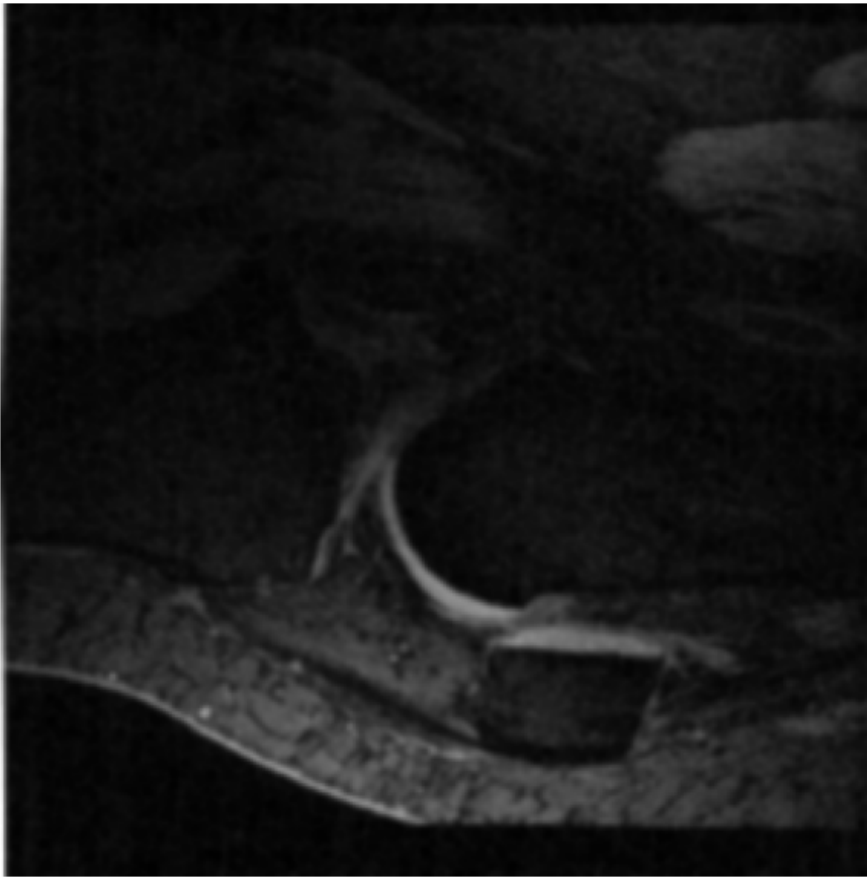


Figure 24: Gaussian denoising applied to the RSS combined image (generated in `notebooks/development.ipynb`).